Introduction to thermodynamics and kinetics of the chemical reactions, radiation, and spectrum.

Thermodynamics/Thermochemistry

THERMODYNAMICS

The branch of science, which deals with energy changes, is known as thermodynamics.

THERMOCHEMISTRY

The branch of chemistry which deals with energy changes in a chemical reaction is called thermochemistry.

WHY HEAT IS EVOLVED OR ABSORBED IN A CHEMICAL REACTION

During chemical reaction, bond breaking and making takes place.

Bond breaking absorbs energy and bond making releases it.

In chemical reaction, energy required to break bonds is not equal to the energy evolved for bond making. Different substances have different energies and thus total energy of reactants is always different from the total energy of products.

This difference of energy is either evolved or absorbed during a chemical reaction.

TYPES OF REACTION

On the basis of absorption or evolution of heat, reactions are classified as

i. ***Exothermic reactions.***
ii. ***Endothermic reactions.***

EXOTHERMIC REACTIONS

The reactions in which heat is evolved are known as exothermic reactions.

In such reactions, total energy of the products is less than the total energy of the reactants. This difference of energy is evolved as heat.

The evolved heat of reaction is indicated by negative sign.

In these reaction, heat is given out and T of the system rises above the room T. After some time heat is lost to the surroundings and T falls to room T.

Examples:

i. $C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$ $\Delta H = -393.5$ KJ/mol

ii. $H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow 2H_2O_{(l)}$ $\Delta H = -285.58$ KJ/mol

iii. $N_{2(g)} + 3H_{2(g)} \longrightarrow 2NH_{3(g)}$ $\Delta H = -41.6$ KJ/mol

Endothermic Reacions

The reactions in which heat is absorbed are known as endothermic reactions.

In such reactions total energy of the products is greater than the total energy of the reactants. This difference of energy is absorbed as heat.

The absorbed heat of reaction is indicated by positive sign.

In these reactions, heat require for reaction is obtained from reactants and thus T of the system falls below the room T. After some time heat is absorbed from surrounding and T again rises to room T.

Examples:

$H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)}$ $\Delta H = +52.96$ KJ/mol

$N_{2(s)} + O_{2(g)} \longrightarrow 2NO_{(g)}$ $\Delta H = +180.51$ KJ/mol

Importance And Limitation Of Thermochemistry

Thermochemistry tells about the energy or heat contents of the system including reactions. This knowledge is used to explain chemical bonding and chemical equilibrium.

However, heat of reaction can be accurately determined only for few reactions. Thus It limits the scope of thermochemistry.

SYSTEM & SURROUNDING

SYSTEM.

Any real or imaginary part of the universe that is under study is called system.

A chemical system is usually a substance undergoing a chemical charge.

SURROUNDING

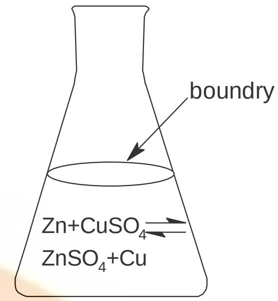
Everything that is not part of the system is called surrounding.

BOUNDRY

The real or imaginary surface separating the system from surrounding is called boundry.

EXAMPLES:

1. ***In experimental work, a specific amount of one or more substances forms a system e.g. 1 mole of O_2 in a cylinder fitted with a piston is a system. The piston, the cylinder and all other objects outside cylinder are surroundings.***
2. ***A cup of water is a system and the air around it and the table on which it is present are all surroundings.***
3. ***In reaction of Zn with $CuSO_4$ solution, it is a system, while flask, air etc are surroundings.***



STATE

The condition of a system is called a state

When a process occurs, the state of system is changed.

Explanation

Consider a water system at a given T and V. This initial condition of water is called initial state. Now if water is heated in a beaker, its condition will be changed. The final condition of the system is called final state.

A comparison of final and initial state tells us about the change taking place in a system.

e.g. Let T_1 and T_2 are the temperatures at initial and final state of the water system then change in temperature will be

$$\Delta T = T_2 - T_1$$

STATE FUNCTION

It is a macroscopic property of system which has definite values for initial and final states and it is independent of the path through which change take place.

Conventionally, capital letters are used to represent a state function.

e.g. Volume (V), Temperature (T), Enthalpy (H), Internal Energy (E) etc.

Example

let initial volume of gas is V_1 . Its volume can be changed to V_2 by changing T or P. The change in volume is given by

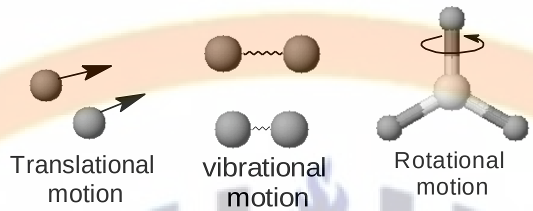
$$\Delta V = V_2 - V_1$$

Since V is a state function, therefore, ΔV is independent of the path through which change takes place (i.e. by changing T or P)

INERNAL ENERGY OF A SYSTEM

It is the sum of all the energies of all the particles (i.e, atom, ions and molecules) within a system.

It depends upon the motion of molecules (i.e, translational, vibrational, rotational) and their intermolecular and intramolecular forces.



In other words it is the total energy of a system due to kinetic energies of its particles plus the potential energies due to binding forces between the particles.

It is denoted by 'E' .

Internal energy is a state function i.e, it depends only on the initial and final state of a system and not on the path through which change takes place.

The internal energy of a system cannot be determined. Only changes in internal energy, denoted by ΔE, can be determined

$$\text{i.e,} \quad \Delta E = E_2 - E_1$$

ENERGY TRANSFER BETWEEN SYSTEM AND SURROUNDINGS.

WORK AND HEAT

When a system undergoes a change, energy is transferred into or out of the system in the form of heat and mechanical work.

WORK

Work is a form of energy and is defined as the product of force and distance.

$$\text{Work} = \text{Force} \times \text{Distance}$$

It is denoted by w

It is not the property of system. Hence it is path function, it is not a state function.

Sign Convention for w

Work done by the system is negative i.e. – w

Work done on the system is positive i.e. +w

Units of w

The SI units of work is Joule (J).

Other unit is erg

$$1 \text{ erg} = 10^{-7} \text{ J}$$

There are many types of work.

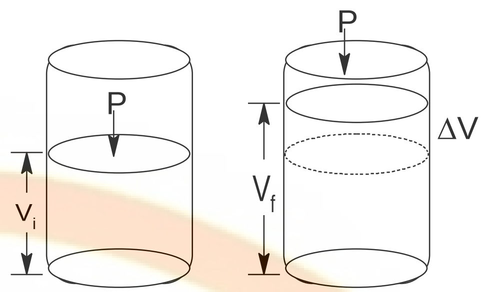
In chemistry most common type of work is pressure-volume work given by

$$w = P \Delta V$$

Where

P = External pressure

ΔV= Change in volume



Expansion work
(P-V work)

HEAT

It is the quantity of energy that flows across the boundry of a system during a change in state, due to the difference in T between system and surroundings.

It is denoted by q.

It is not the property of system. Hence it is path function, it is not a state function.

Sign convention for q.

Heat gained by the system is positive i.e. +q

Heat lost by the system is negative i.e. – q

Units of q.

The SI unit of heat is joule (J).

Other unit is calorie (cal).

1 cal = 4.184 J

FIRST LAW OF THERMODYNAMICS

The law of conservation of energy is also called the first law of thermodynamics.

It states.

"Energy can neither be created nor destroyed although it can be transformed form one form to another ".

OR it can also be stated as

"The energy of the system and surrounding is conserved".

Explanation

Energy is transformed into & out of a system in the form of heat and work.

If some heat is added to a system & some work is done on it then its internal energy is changed. The energy change is equal to the sum of both heat & work so that total energy of the system & surroundings remain constant.

i.e; $\Delta E = q + w$ ***------ (1)***

This eq is called the mathematical form of first law of thermodynamics.

In this eq

ΔE ***= change in internal energy of system.***

q ***= heat added to the system;***

w ***= work done on the system.***

Sign Conventions

In eq (1) following sign conventions are used.

ΔE ***is negative when system loses energy i.e;*** $-\Delta E$

ΔE ***is positive when system gains energy i.e;*** $+\Delta E$

Work done on the system in positive i.e; $+ w$

Work done by the system is negative i.e; $- w$

Heat gained by the system is positive i.e; $+ q$

Heat lost by the system is negative i.e; $- q$

Both heat added to and work done on the system increases the internal energy of a system therefore both these are taken positive.

While both heat given out or work done by a system decreases the internal energy of a system therefore both these are taken negative.

HEAT CHANGES AT CONSTANT VOLUME

At constant volume, let the heat supplied to the system is 'q_v***' then first law of thermodynamics can be written as***

$$\Delta E = q_v + w$$

$$\text{Since } w = -P\,\Delta V$$

$$\Delta E = q_v - P\,\Delta V$$

Since volume is constant, therefore , $\Delta V = 0$

Thus $\Delta E = q_v$

It means that heat exchanged at constant volume is used to change the internal energy of the system and no work is done.

HEAT CHANGES AT CONSTANT PRESSURE (ENTHALPY)

Enthalpy can be defined as

" Enthalpy of a system is the sum of its internal energy and PV work, and it represents the total heat content of a system."

Mathematically

$$H = E + PV$$

Explanation

At constant P, when heat is supplied to the system. A part of this heat is used to increase the internal energy of a system and a part is used to do P-V work. Thus enthalpy is used to represent the change in state of the system.

Enthalpy is also called total heat content of the system.

Properties

1. ***It is a state function***
2. ***Its unit is Joule (J)***
3. ***It can not be determined for a system in a given state. Only change in enthalpy can be determined.***

Since enthalpy is

$$H = E + PV$$

Change in enthalpy is given by

$$\Delta H = \Delta E + \Delta (PV)$$

$$\Delta H = \Delta E + V\Delta P + P\Delta V$$

At constant pressure, $\Delta P = 0$

$$\Delta H = \Delta E + P\Delta V \quad \_\_\_ \ (1)$$

According to first law of thermodynamics at constant P

$$\Delta E = q_p + w$$

Since $w = -P\Delta V$

$$\Delta E = q_p - P\Delta V \_\_\_\_ (2)$$

Put eq (2) in eq (1)

$$\Delta H = q_p - P\Delta V + P\Delta V$$

$$\Delta H = q_p$$

It means that heat exchanged at constant pressure is used to change the enthalpy of a system.

Relationship Between ΔH ***&*** ΔE

Change in Enthalpy is given by

$$\Delta H = \Delta E + P\Delta V \_\_\_\_ (5)$$

For reactions involving only solids and liquids, change in volume is zero i.e. $\Delta V = 0$

Hence $\Delta H \cong \Delta E$

STANDARAD STATE

The natural physical state of a substance at 250C (298K) and 1 atm. pressure is known as standard state of the substance e,g;

The standard state of CO_2 ***is gas***

The standard state of H_2O ***is liquid.***

The standard state of Fe is solid.

ENTHALPY OF REACTION (ΔH_r^o)

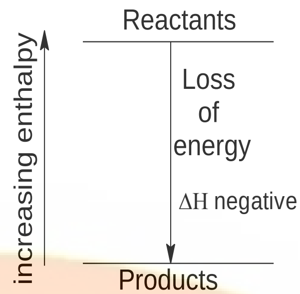
The enthalpy change which occurs when the no. of moles of reactants as indicated by the balanced chemical equation react together completely to give products under standard conditions.

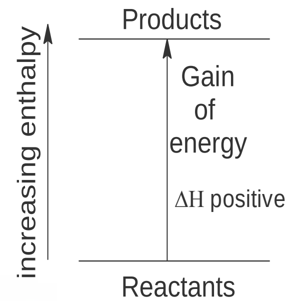
All the reactants and products are in their standard physical states.

Note

- ***In exothermic reactions, enthalpy of product is less than enthalpy of reactant. Hence heat is evolved and ΔH_r^o is negative.e.g.***

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \quad \Delta H = -393.5 \text{ KJ/mol}$$





- ***In endothermic reactions, enthalpy of product is greater than enthalpy of reactants. Hence heat is absorbed and ΔH_r^o is positive.***

<u>ENTHALPY OF FORMATION</u> (ΔH_f^o)

<u>The heat of reaction when 1 mole of the compound is formed from its element, in their standard state is known as standard heat of formation of the compound.</u>

<u>Examples</u>

$$Mg_{(s)} + \tfrac{1}{2} O_{2\,(g)} \longrightarrow MgO_{(s)} \qquad \Delta H_f^o = -692 \text{ KJ/mol}$$

$$C_{(s)} + O_{2\,(g)} \longrightarrow CO_{2\,(g)} \qquad \Delta H_f^o = -393.7 \text{ KJ/mol}$$

<u>ENTHALPY OF ATOMIZATION</u> (ΔH_{at}^o)

<u>The enthalpy change when one mole of gaseous atoms are formed from the elements under standard conditions is called enthalpy of atomization of an element.</u>

<u>Example</u>

$$\tfrac{1}{2} H_{2\,(g)} \longrightarrow H_{(g)} \qquad \Delta H_{at}^o = 218 \text{ KJ/mol}$$

Enthalpies of atomization can be determined by various methods.

<u>ENTHALPY OF NEUTRALIZATION</u> (ΔH_n^o)

<u>The amount of heat evolved when one mole of hydrogen ions (H^+), from an acid, react with one mole of hydroxide ions (OH^-), from a base to form one mole of water is called enthalpy of neutralization</u>

<u>Example</u>

The enthalpy of neutralization of NaOH with HCl is –57.4 KJ/mol

HCl is a strong acid and NaOH is a strong base

Both ionizes in water completely

$$HCl_{(aq)} \longrightarrow H^+_{(aq)} + Cl^-_{(aq)}$$

$$NaOH_{(aq)} \longrightarrow Na^+_{(aq)} + OH^-_{(aq)}$$

On mixing these solutions. H^+ and OH^- ions react together to form water and heat is evolved. While Na^+ and Cl^- ions are set free in such solution.

Thus heat released during this process is actually the heat of formation of H_2O

$$Na^+_{(aq)} + OH^-_{(aq)} + H^+_{(aq)} + Cl^-_{(aq)} \longrightarrow H_2O + Na^+_{(aq)} + Cl^-_{(aq)}$$

Or $$H^+_{(aq)} + OH^-_{(aq)} \longrightarrow H_2O_{(l)} \qquad \Delta H^o_n = -57.4\ KJ/mol$$

Note:

Enthalpy of neutralization of a strong acid with strong base is always approximately constant i.e. – 57.4 KJ/mol

ENTHALPY OF COMBUSTION (ΔH^o_c)

The enthalpy change when one mole of a substance is completely burnt in excess of oxygen under standard conditions is called enthalpy of combustion of a substance.

Example

$$C_2H_5OH_{(l)} + 3O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O_{(l)} \quad \Delta H^o_c = -1368\ KJ/mol$$

ENTHALPY OF SOLUTION(ΔH^o_{sol})

The amount of heat absorbed or evolved when one mole of a substance is dissolved in so much solvent that further dilution results in no detectable heat change is called enthalpy of solution of substance.

Examples

1. *Enthalpy of solution of NH_4Cl is +15.1 KJ/mol.*
   During this heat is absorbed from surrounding and solvent is cooled. It is an endothermic process.

2. *Enthalpy of solution of Na_2CO_3 is –25 KJ/mol.*
   During this heat is released and T of the solvent rises. It is an exothermic process.

MEASUREMENT OF ENTHALPIES

BOMB CALORIMETER

It is used for accurate determination of heat of combustion of food, fuel and other substances.

Construction

It consists of a cylindrical steel vessel (steel bomb),

lined with enamel to prevent corrosion

and a screw valve at the top.

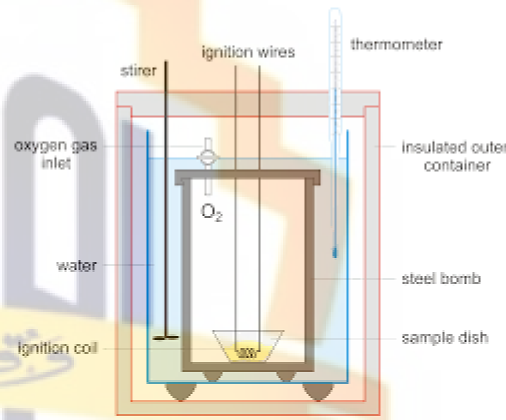
A platinum crucible is present

inside steel bomb.

There is also present an

electrical ignition coil.

The bomb is immersed in a



known mass of H_2O in an insulated calorimeter as shown in fig. The T is measured with a thermometer.

Working

A known mass of substance is placed on a platinum crucible. The initial T of water is noted. The substance is then ignited electrically by ignition coil.

The water is stirred and its T is noted continuously after every 30 sec.

The maximum T is noted from thermometer.

*The difference of initial and final T gives the change in T (Δ**T**)*

If no. of moles of a substance burnt is 'm' and specific heat of whole system (calorimeter etc) is 's' , then heat evolved or absorbed 'q' is determined by the formula.

$$q = m \times s \times \Delta T$$

THEREMOCHEMICAL LAWS

There are two laws of thermochemistry

1. <u>FIRST LAW OF THERMOHEMISTRY</u>

It states

<u>The heat of decomposition of a compound is equal to its heat of formation but with opposite sign.</u>

e.g; The heat of formation of one mole of CO_2 under standard conditions is

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H = -393.7\ KJ\ mol^{-1}$$

The same amount of heat is required to decompose one mole of $CO_{2(g)}$ into $C_{(s)}$ & $O_{2(g)}$.

$$CO_{2(g)} \longrightarrow C_{(s)} + O_{2(g)} \qquad \Delta H = +393.51\ KJ\ mol^{-1}$$

2. <u>Second Law Of Thermochemistry Or Hesse's Law</u>

G.H. Hess experimentally discovered it in 1840.

It states

"*<u>The amount of heat evolved or observed in a process, including a chemical change, is the same whether the process takes place in one step or in several steps.</u>*

Or

<u>If a chemical reaction takes place by different ways, the net energy change is same, regardless of the route by which the chemical change occurs, provided the initial and final states are the same.</u>

$$\begin{tikzcd} A \arrow[r, "\triangle H_4"] \arrow[d, "\triangle H_1"'] & B \\ C \arrow[r, "\triangle H_2"'] & D \arrow[u, "\triangle H_3"'] \end{tikzcd}$$

Mathematically from fig

$$\Delta H_1 + \Delta H_2 + \Delta H_3 = \Delta H_4$$

Thus if system returns to its initial state in a cyclic process, we can write

$$\sum \Delta H_{(cycle)} = 0$$

<u>Hence net enthalpy change in a closed cycle is always equal to zero, if the initial and final states are same.</u>

<u>Importance And Examples (Or Proof)</u>

There are several processes for which ΔH can not be determined directly.

e.g. ΔH_f^o of Al_2O_3 and B_2O_3 can not be determined. It is because, a protective layer is present over the surface of element that resist in complete burning.

For such reaction heat of reaction is determined indirectly by using Hess's law

Similarly ΔH_f^o of CO can not be measured directly due to production of CO_2 along with it. It can be determined in following ways.

<u>Example 1:</u>

<u>(Determination of ΔH_f^o for $CO_{(g)}$)</u>

Combustion of carbon can occurs in two ways.

<u>First way</u>

When C is burned in excess of O_2, it is directly converted into CO_2.

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H_1 = -393.51\ KJ\ mol^{-1}$$

<u>Second way</u>

When C is burned in limited supply of oxygen CO is formed.
This CO on further oxidation gives CO_2

$$C_{(s)} + \tfrac{1}{2} O_{2(g)} \longrightarrow CO_{(g)} \qquad \Delta H_2 = ?$$

$$CO_{(s)} + \tfrac{1}{2} O_{2(g)} \longrightarrow CO_{2(g)} \qquad \Delta H_3 = -282.98\ KJ\ mol^{-1}$$

According to Hess's law

$$\Delta H_1 = \Delta H_2 + \Delta H_3$$

or $\quad \Delta H_2 = \Delta H_1 - \Delta H_3$

$\Delta H_2 = -393.7 - (-283)$

$\Delta H_2 = -110$ *KJ*

Hence heat of formation of $CO_{(g)}$ from $C_{(s)}$ is -110 KJ mol^{-1}.

<u>THE BORN-HABER CYCLE</u>

It is based on the principle that

<u>*"The sum of energy changes for a closed cyclic process is zero, if the initial and final states are same"*</u>

Born-Haber cycle is used to determine lattice energy of ionic crystals

Lattice energy is defined as

<u>***"The amount of energy released when gaseous ions of opposite charges, combine to give one mole of a crystalline ionic compound."***</u>

Lattice energy can not be determined directly

It is determined by a closed Born-Haber cycle

<u>***Example***</u>

<u>***(Determination of lattice energy of NaCl)***</u>

$$\begin{tikzcd}
Na_{(s)} + 1/2\, Cl_{2(g)} \arrow[r, "\Delta H_x"] \arrow[d, "\Delta H^0_f"'] & Na^+_{(g)} + Cl^-_{(g)} \arrow[dl, "\Delta H^0_{latt}"] \\
Na^+Cl^-_{(s)} &
\end{tikzcd}$$

Consider the case of NaCl

Following closed cycle can be drawn for NaCl.

From closed triangle, according to Hess's law, we can write

$$\Delta H^o_f = \Delta H^o_{latt} + \Delta H_x$$

$$\text{or} \quad \Delta H^o_{latt} = \Delta H^o_f - \Delta H_x \quad \_\_\_\_ \ (1)$$

Thus if ΔH^o_f and ΔH_x are known, then ΔH^o_{latt} can be calculated

ΔH^o_f of NaCl can be measured in a calorimeter and its value is -411 KJ mol^{-1}

$$\Delta H^o_f = -411 \text{ KJ mol}^{-1} \quad \_\_\_ \ (2)$$

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ΔH_x can be calculated as follow

The overall process is

$$Na_{(s)} + \tfrac{1}{2} Cl_2 \longrightarrow Na^+_{(g)} + Cl^-_{(g)} \qquad \Delta H_x = ?$$

This process involves following steps

1. ***Atomization of $Na_{(s)}$ to $Na_{(g)}$.***
   Energy required for this process is determined from values of its heat of fusion, heat of vaporization and specific heat capacity.

$$Na_{(s)} \longrightarrow Na_{(g)} \qquad \Delta H_{at} = 108 \text{ KJ mol}^{-1}$$

2. ***Conversion of $Na_{(s)}$ to $Na_{(g)}$ ion.***
   This requires ionization energy of $Na_{(g)}$ which can be determined by spectroscopy.

$$Na_{(g)} \longrightarrow Na^+_{(g)} + e^- \qquad \Delta H_i = 500 \text{ KJ mol}^{-1}$$

3. ***Atomization of $Cl_{2(g)}$ to $Cl_{(g)}$.***
   Energy required for this process is determined by spectroscopy.

$$\tfrac{1}{2} Cl_{2(g)} \longrightarrow 2\, Cl_{(g)} \qquad \Delta H_{at} = 500 \text{ KJ mol}^{-1}$$

4. ***Conversion of $Cl_{(g)}$ to $Cl^-_{(g)}$.***
   This process is the electron affinity of Cl. Its value can also be determined experimently.

$$Cl_{(g)} + e^- \longrightarrow Cl^-_{(g)} \qquad \Delta H_e = -364 \text{ KJ mol}^{-1}$$

Hence

$$\Delta H_x = \Delta H_{at}(Na) + \Delta H_i(Na) + \Delta H_{at}(Cl_2) + \Delta H_e(Na)$$

$$\Delta H_x = 108 + 500 + 121 - 364$$

$$\Delta H_x = 365 \text{ KJ mol}^{-1} \quad \_\_\_ \ (3)$$

Put values of ΔH_x from eq (3) and ΔH^o_f from eq (2) in eq (1) , we get

$$\Delta H^o_{latt} = \Delta H^o_f - \Delta H_x$$

$$\Delta H^o_{latt} = -411 - (365) = -776 \text{ KJ mol}^{-1}$$

Hence lattice energy of NaCl is –776 KJ mol^{-1}. It represents the forces of attraction in NaCl crystals. Lattice energies are used to explain structure, bonding and properties of ionic compounds.

REACTION KINETICS

CHEMICAL KINETICS

The branch of chemistry, which deals with the study of reaction rates and the factors, which affect these rates, is known as chemical kinetics.

Reaction kinetics help to determine the mechanism of reactions.

TYPES OF CHEMICAL REACTIONS

On the basis of velocity of reactions, there are three types of reactions.

Slow Reactions

Some reactions occur very slowly and take even years to complete.

e.g. Consider the reaction

$$2H_2 + O_2 \longrightarrow 2H_2O$$

This reaction occurs very slowly and might take, perhaps, decades to complete.

Similar are the rusting of iron, chemical weathering of stone, fermentation of sugars etc.

Fast Reactions

Some reactions take almost no time to complete. Such reactions are fast reactions.

e.g. Consider the reaction

$$AgNO_{3(aq)} + NaCl_{(aq)} \longrightarrow NaNO_{3(aq)} + AgCl_{(s)}$$

This reaction is completed as soon as $AgNO_{3(aq)}$ and $NaCl_{(aq)}$ touches each other.

Moderate Reactions

Some reactions occur at moderate rates, neither too slow nor too fast.

e.g. Hydrolysis of ester

$$CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$$

MULTI-STEP REACTIONS

All reactions occur in single step or many steps.

For multistep reactions. One of the step is slow. This is called rate determining step. Other steps do not affect the rate of reaction.

RATE OF REACTION

It is the change in concentration of reactants or products in a unit time

$$Rate = \frac{Change\ in\ conc.}{Change\ in\ time} = \frac{\Delta c}{\Delta t}$$

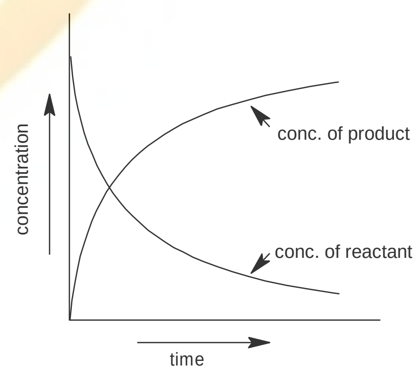
As the reaction proceeds, the concentration of reactant decreases with time and the concentration of product increases with time. Therefore, rate of reaction can be defined as

The rate of decrease in concentration of reactants

Or *The rate of increase in concentration of products.*

Explanation

When reaction starts, conc. of reactants decreases with time, while concentration of product increases. At start, rate of reaction is generally very fast. Thus the slope of curve is steepest for both reactants and products as shown in the fig.



After some time the slope becomes less steep showing that reaction rate decreases with time.

Units

Usually, concentrations are expressed in mol dm^{-3} and time in sec. Therefore, units of rate are

$$Rate = \frac{\Delta c}{\Delta t} = \frac{mol\ /\ dm^3}{sec} = mol\ dm^{-3}\ sec^{-1}$$

<u>For gaseous reactions, partial pressures of gases are used instead of molar concentrations.</u>

<u>AVERAGE RATE</u>

<u>The rate or reaction between specific time intervals is called the average rate of reaction.</u>

Let amount of product formed at t_1 is c_1 and at t_2 is c_2 , then

$$\textit{Average rate} = \frac{c_2 - c_1}{t_2 - t_1} = \frac{\Delta c}{\Delta t}$$

<u>INSTANTANEOUS RATE</u>

<u>The rate at any one instant during the interval is called instantaneous rate.</u>

Let 'x' is the amount of product produced at any time 't' then instantaneous rate can be expressed as

$$\textit{Instantaneous Rate} = \frac{dx}{dt}$$

The dx (differential of x w.r.t. 't') is called rate of change of conc. w.r.t. time. i.e. it represents instantaneous reactions rate.

dx and dt are infinitesimally small changes in concentration and time respectively.

For a general reaction

$$A \longrightarrow B$$

The rate of reaction can be expressed in terms of rate of disappearance of A or rate of appearance of B i.e.

$$Rate = -\frac{d[A]}{dt} \ \ or \ + \frac{d[B]}{dt}$$

Where 'd [A]' and 'd [B]' are change in concentrations of A and B respectively.

<u>Negative sign indicates that conc. Of A decreases with time, while</u>

<u>Positive sign indicates that conc. Of B increases with time.</u>

<u>Note</u>

- *At start, instantaneous rate is greater than average rate*
- *At the end of the interval the instantaneous rate becomes less than average rate.*

- *When the time interval approaches zero, average and instantaneous rate and average rates become equal.*

RATE CONSTANT OR VELOCITY CONSTANT OR SPECIFIC RATE CONSTANT

According to law of Mass Action

Rate of a chemical reaction is directly proportional to the product of concentration of reaction substances, each raised to the appropriate power.

Thus for a simple reaction

$$A + B \longrightarrow products$$

$$Rate = k[A][B]$$

Where k = specific rate constant

When [A]=[B]=1 mol/dm^3 then

$$Rate = k$$

Thus rate constant can be defined as

Rate constant is the rate of reaction when the concentrations of reactants are unity

ORDER OF REACTION

The number of reacting molecules whose concentration changes during a reaction is called order of reaction

Consider a general reaction

$$aA + bB \longrightarrow products$$

Rate eq. For this reaction will be

$$Rate\ of\ reaction = k[A]^m[B]^n$$

m and n may or may not be equal to a & b respectively.

m and n may be either whole number or in fraction or zero

m is the order of reaction w.r.t A

n is the order of reaction w.r.t. B

The sum m + n is called overall order of reaction

Thus order of reaction can be defined as

The sum of the powers to which concentrations are raised in the rate law expression is called order of reaction

In chemical kinetics, reactions are classified as zero order, first order, second order, third order.

Zero Order Reaction

When the reaction is independent of the concentration of reactants, it is called zero order reaction.

Example

Decomposition of HI (on gold surface)

$$2HI \xrightarrow{\text{on gold surface}} H_2 + I_2$$

The experimental rate equation for this reaction is

$$\text{Rate} = k[HI]^0 = k$$

This reaction is independent of the conc. of HI, hence it is a zero order reaction.

First Order Reactions

When sum of powers to which concentrations are raised in the rate law expression is equal to 1.

Examples

1. Decomposition of nitrogen pentoxide.

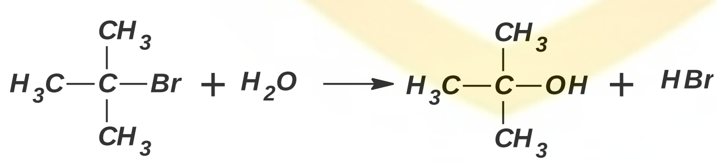
$$2N_2O_{5\,(g)} \longrightarrow 2N_2O_{4\,(g)} + O_{2\,(g)}$$

The experimental rate equation for this reaction is

$$\text{Rate} = k[N_2O_5]$$

Hence it is a first order w.r.t. the concentration of N_2O_5.

2. *Hydrolysis of tertiary butyl bromide*



tertiary butyl bromide   tertiary butyl alcohol

The experimental rate equation for this reaction is

$$\text{Rate} = k\,[(CH_3)_3CBr]$$

Hence it is a first order w.r.t. the concentration of $(CH_3)_3CBr$.

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The rate of reaction is independent of the concentration of water because, it is present in very large excess. Such types of reactions are called pseudo first order reactions.

Second Order Reactions

When sum of powers to which concentrations are raised in the rate law expression is equal to 2.

Example

Oxidation of nitric oxide with ozone

$$NO_{(g)} + O_{3\,(g)} \longrightarrow NO_{2\,(g)} + O_{2\,(g)}$$

The experimental rate equation for this reaction is

$$Rate = k\,[NO][O_3]$$

This reaction is first order w.r.t. NO and first order w.r.t. O_3. The sum of the individual orders is equal to two. Hence it is second order reaction.

Third Order Reactions

When sum of powers to which concentrations are raised in the rate law expression is equal to

Example

Reaction of $FeCl_3$ with KI

$$2FeCl_{3\,(aq)} + 6KI_{(aq)} \longrightarrow 2FeCl_{2\,(aq)} + 6KCl_{(aq)} + I_2$$

This reaction involves eight reactant molecules But the experimental rate equation for this reaction is

$$Rate = k\,[FeCl_3]\,[KI]^2$$

This reaction is first order w.r.t. $FeCl_3$ and second order w.r.t. KI.

The sum of the individual orders is equal to three. Hence it is a third order reaction.

It shows that it is a multistep reaction and one step is the rate determining step. The possible steps are

$$2FeCl_{3\,(aq)} + 4KI_{(aq)} \longrightarrow 2FeI_{2\,(aq)} + 4KCl_{(aq)} + 2Cl^{-}_{(aq)}$$
$$2KI_{(aq)} + 2Cl^{-}_{(aq)} \longrightarrow 2KCl_{(aq)} + I_{2\,(s)}$$

Note

The order of a reaction is usually positive integer or a zero, but it can also be in fraction or can have a negative value.

Formation Of Carbon Tetrachloride Form Chloroform.

$$CHCl_3\,(l) + Cl_2\,(g) \longrightarrow CCl_4\,(l) + HCl\,(g)$$

The experimental rate equation for this reaction is

$$Rate = k\,[CHCl_3][Cl2]^{1/2}$$

The sum of exponents will be 1 + ½ = 1.5

Hence the order of this reaction is 1.5.

Decomposition of Ozone

$$2O_3\,(g) \longrightarrow 3O_2\,(g)$$

The experimental rate equation for this reaction is

$$Rate = k\,[O_3]^2\,[O_2]^{-1}$$

It shows that during reaction, when the concentration of oxygen doubles, the rate becomes half.

RATE LAW

The experimental relationship between a reaction rate and the concentration of the reactants is known as the rate law or the rate equation for that reaction.

Consider a general reaction

$$aA + bB \longrightarrow Products$$

Rate eq. for this reaction will be

$$Rate = k\,[A]^m\,[B]^n$$

m and n may or may not be equal to a & b respectively.

m and n may be either whole number or in fraction or zero.

m and n can only be determined experimentally and not theoretically.

Rate law expression is, therefore, an experimental expression.

HALF-LIFE PERIOD OR HALF LIFE TIME

The time required to convert 50% or the reactants into products is called half life of the reaction.

It is denoted by $t_{1/2}$ or $t_{0.5}$

Example 1:

Decmoposition of N_2O_5

$$2N_2O_{5\,(g)} \longrightarrow 2N_2O_{4\,(g)} + O_{2\,(g)}$$

Half life period for decomposition is N_2O_5 is 24 minutes at 45^0C

It means that if reaction is started with 0.1 mol/dm^3 of N_2O_5 then after 24 minutes, 0.05 mole/dm^3 will be left behind.

Similarly, after 48 minutes (two half times), 0.025 mol/dm^3 will be left behind and after 72 minutes (three half times), 0.0125 mol/dm^3 will be left behind.

Decomposition of N_2O_5 is a first order reaction. Above experiment clearly shows that half life time of a first order reaction is independent of the initial concentration of reactants.

Example 2

Disintegration of $^{235}_{92}U$

It has a half life period of 7.1×10^8 years or 710 million years. It is also a first order reaction.

It means that if 1 Kg of $^{235}_{92}U$ is present, then after 710 million years, 0.5 Kg is converted into daughter elements. And in the next 710 years, 0.25 Kg of the $^{235}_{92}U$ will be consumed.

Relationship Between Half Life Time And Order Of Reaction.

Generally,

Half life period is inversely proportional to initial conc. of reactants raised to the power one less than the order of reaction.

$$(t_{1/2})_n \;\alpha\; \frac{1}{a^{n-1}}$$

Where n = Order of reaction

a = Initial conc. of reactants

$(t_{1/2})_n$ = Half life period for n^{th} order reaction.

Thus

- ***For first order reaction, half life period is independent of the initial conc. Of reactants.***

$$\text{i.e} \quad (t_{1/2})_1 \ \alpha \ \frac{1}{a^0}$$

Where ($t_{1/2}$)$_1$ is half life period for first order reaction and 'a' is the initial conc.

- ***For Second order reaction, half life period is inversely proportional to the initial conc. of reactants.***

$$\text{i.e} \quad (t_{1/2})_2 \ \alpha \ \frac{1}{a^1}$$

Where ($t_{1/2}$)$_2$ is half life period for second order reaction and 'a' is the initial conc.

- ***For Third order reaction, half life period is inversely proportional to the square of initial conc. of reactants.***

$$\text{i.e} \quad (t_{1/2})_3 \ \alpha \ \frac{1}{a^2}$$

Where ($t_{1/2}$)$_3$ is half life period for third order reaction and 'a' is the initial conc. of reactants.

Hence half-life period can be used to determine the order of reaction.

RATE DETERMINING STEP

The reaction, which determines the overall rate of reaction, is called rate-determining step.

For multistep reaction, rate determining step is the slowest step.

Explanation

All reactions occur in single step or many steps.

For single step reaction, one single step is always the rate determining step.

For multistep reactions. One of the step is slowest. This is called rate determining step. Other steps do not affect the rate of reaction.

<u>The total number of reactant molecules that takes part in rate – determining step appear in the experimental rate equations of reaction.</u>

Hence experimental rate equations can give information about mechanism of reactions.

<u>Example 1</u>

<u>Reaction of NO_2 with CO</u>

$$NO_{2(g)} + CO_{(g)} \longrightarrow NO_{(g)} + CO_{2(g)}$$

The experimental rate equation for this reaction is

Rate = k $[NO_2]^2$

This rate equation shows that, rate is independent of conc. of CO. Thus it shows that it is a multistep reaction. And one step is slow which is the rate-determining step and involves two molecules of NO_2.

Thus following mechanism can be written

$$NO_2 + NO_2 \xrightarrow{slow} NO_3 + NO$$

$$NO_3 + CO \xrightarrow{fast} NO_2 + CO_2$$

Thus first step is slow and it is the rate determining step

NO_3 is not produced in the reaction. It is consumed as soon as it is formed. Such a specie is called reaction intermediate.

<u>The specie which produce only for short period of time during a reaction is called reaction intermediate.</u>

<u>Reaction intermediate is unstable as compare to both reactants and products</u>.

<u>In some cases, under certain conditions, it may be stable and can be isolated.</u>

<u>Example 2</u>

<u>Reaction of $FeCl_3$ with KI</u>

$$2FeCl_{3\,(aq)} + 6KI_{(aq)} \longrightarrow 2FeCl_{2\,(aq)} + 6KCl_{(aq)} + I_2$$

This reaction involves eight reactant molecules But the experimental rate equation for this reaction is

$$\text{Rate} = k\,[FeCl_3]\,[KI]^2$$

It is a third order reaction.

Thus it shows that it is a multistep reaction. The possible steps are

$2FeCl_{3\,(aq)} + 4KI_{(aq)} \longrightarrow 2FeI_{2\,(aq)} + 4KCl_{(aq)} + 2Cl^{-}_{(aq)}$

$\boldsymbol{2KI_{(aq)} + 2Cl^{-}_{(aq)} \longrightarrow 2KCl_{(aq)} + I_{2\,(s)}}$

ENERGY OF ACTIVATION

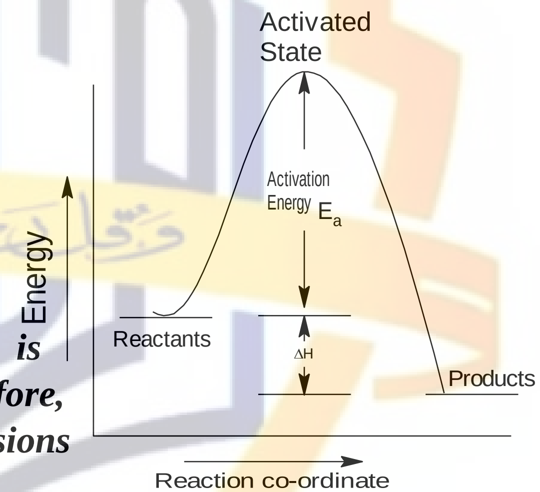
The minimum amount of energy which molecules must have, in addition to their average K.E., to form an activated complex, is called the Activation Energy.

It is denoted by E_a

Units

It is expressed in joules

Explanation



In a reaction system, molecules constantly collide with one another.

At ordinary temperature, activation energy is usually very high for any reaction system. Therefore, most of the molecules simply rebound after collisions and do not form activated complex.

However, if some of the molecules gain enough energy after collisions to form activated complex, they will start the reaction. Such collisions, which form an activated complex are called effective collisions. For effective collisions, molecules must have proper orientation.

The number of effective collisions increases with increase in T and thus rate of reaction is also increased and vice versa.

Colliding molecule first form a high energy transition state or Activated Complex, which then leads to product.

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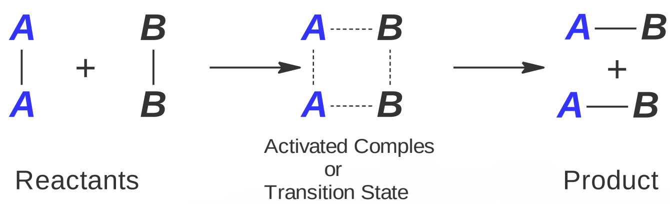
It can be represented as



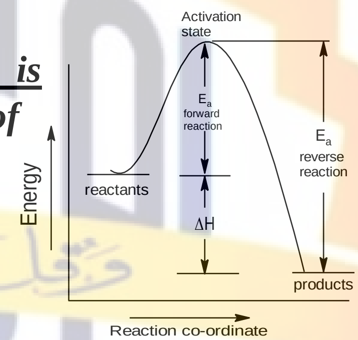
Activated complex is an unstable, short-lived intermediate. It at once changes into product.

RELATIONSHIP BETWEEN ΔH AND Ea

Exothermic reaction

For exothermic reactions, energy of reactant is higher than the energy of product. ***The difference of energy is released as heat.***

It has been shown in the fig.

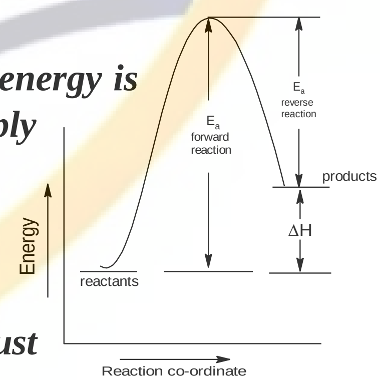


Exothermic Reactions

Endothermic reactions

For endothermic reactions, energy of reactant is lower than the energy of product. ***The difference of energy is absorbed as heat. For such reactions a continuous supply of energy is needed.***

It has been shown in the fig.



Endothermic Reactions

It is clear that, both for exothermic and endothermic reactions, there is energy barrier, which must be overcome in order to start the reaction.

For all reactions, activation energy is different for forward and backward reactions. For exothermic reaction, E_a is less for forward reaction than backward reaction. And for endothermic reaction, E_a is more for forward than backward reaction.

E_a give information about mechanism of reaction.

FACTORS AFFECTING REACTION RATES

Following factors affect the reaction rates

1. ***Nature of reactant***
2. ***Concentration of reactants***
3. ***Temperature***
4. ***Catalyst***
5. ***Surface Area***
6. ***Light***

1. NATURE OF THE REACTANT

Reaction rates are greatly affected by the nature of Reactant.

The reactivity of a substance is controlled by its electronic arrangement.

Examples

- ***Elements of group I–A reacts more rapidly with H_2O than elements of group II –A.***
- ***Neutralization and double decomposition reactions are very fast***
- ***Ionic reactions are very fast e.g. reaction between HCl and NaOH is completed in just 10^{-6} sec, at room T.***
- ***Redox reactions involve transfer of electrons, therefore, these are slower than ionic reactions.***

2. EFFECT OF CONCENTRATION

Rate of reaction is affected by changing conc. in accordance with law of Mass Action. This law states

" The rate at which a substance reacts, is directly proportional to its active mass, and the rate of reaction is directly proportional to the product of active masses of reacting substances".

Thus by increasing conc. of any reactant, rate of reaction is increased and by decreasing conc. rate of reaction is decreased.

Examples

- ***Combustion in normal air (with 21 % O_2) becomes faster in pure O_2***
- ***Lime stone reacts with different rate with different conc. of HCl***

- ***Conc. of gases can be increased by increasing their pressure and hence rate of reaction increases.***
- ***Consider the reaction***

$$2H_2 + 2NO \longrightarrow 2H_2O + N_2$$

Experimental data for this reaction is given below

NO (mol/dm^3)	0.006	0.006	0.006	0.001	0.002	0.003
H_2 (mol/dm^3)	0.001	0.002	0.003	0.009	0.009	0.009
Initial rate (atm/min)	0.025	0.050	0.075	0.0063	0.025	0.056

Experiments show, that by keeping conc. of NO constant, and doubling the conc. of H_2 , doubles the reaction rate and tripling the conc. of H_2 triples the rate.

Thus rate of reaction depends on the first power of conc. of H_2

i.e. $\quad \text{Rate} \propto [H_2]$ _____ (1)

Similarly, keeping conc. of H_2 constant and doubling conc. of NO, raises the rate to 4 times, and tripling conc. of NO, raises the rate to 9 times.

Thus rate of reaction depends on square of conc. of NO

i.e. $\quad \text{Rate} \propto [NO]^2$ ____ (2)

Combining (1) and (2)

$$\text{Rate} \propto [H_2]\,[NO]^2$$

$$\text{Rate} = k\,[H_2]\,[NO]^2$$

It's the rate law for the given reaction. It shows that it is a third order reaction. Thus rate eq for this reaction is not according to balanced equation.

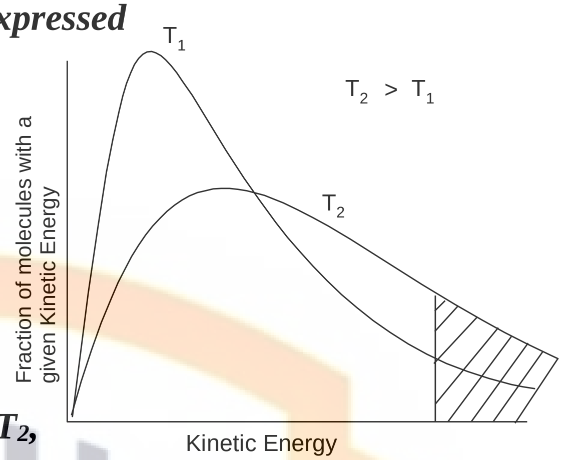
To satisfy rate eq, following mechanism can be proposed

$$2NO + H_2 \xrightarrow{slow} N_2 + H_2O_2 \qquad \text{(rate determining step)}$$

$$H_2O_2 + H_2 \xrightarrow{fast} 2H_2O$$

3. <u>EFFECT OF TEMPERATURE</u>

Effect of T on reaction rate can be expressed graphically.



To start reaction, molecules must gain activation energy E_a

A low T mostly molecules posses average energy and only few molecules posses necessary E_a for reaction.

Graph shows, that at higher temperature T_2, there are more molecules having K.E. equal to E_a, than at low temperature T_1. Thus rate of reaction is higher at T_2 than at T_1.

***Arrhenius Equation** (Effect of T)*

Increase in T increases the rate of reaction and vice versa.

Generally, rate of reaction doubles for every 10^0C rise in T.

Effect of T on reaction rate is best explained by Arrhenius eq.

$$k = Ae^{-Ea/RT} \quad \_\_\_\_\_ \quad (1)$$

Where *k= rate constant*

A = Frequency factor or Arrhenius constant

e = Base of natural logarithm

E_a = Energy of Activation

R = General gas constant

T = Absolute temperature

Taking natural log of eq (1)

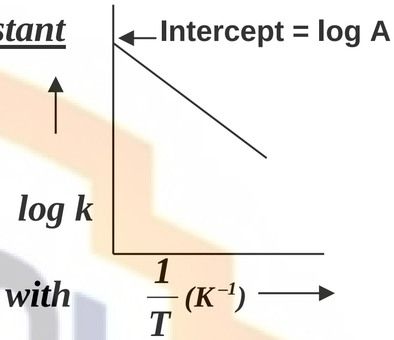
$$\ln k = \ln A - \frac{E_a}{RT} \ln e$$

$$2.303\ \log k = 2.303\ \log A - \frac{E_a}{RT}$$

$$\log k = \log A - \frac{E_a}{2.303\ RT} \qquad \text{\_\_\_}(2)$$

This eq shows that

- ***<u>Higher the activation energy, lower will be rate constant</u>***
- ***<u>Higher the T, higher will be the rate constant</u>***



Eq (2) is an equation of straight line.

A graph between log k and $\frac{1}{T}$ ***, will give a straight line with***

$$\text{Slope} = \frac{-E_a}{2.303\ R} \qquad \text{and intercept} = \log A$$

Thus $$-\text{Slope} = \frac{-E_a}{2.303\ R}$$

or $$E_a = \text{slope} \times 2.303R$$

Radiation chemistry

Radiation chemistry is a branch of chemistry (some say physical chemistry) that studies chemical transformations in materials exposed to high-energy radiations. It uses radiation as the initiator of chemical reactions, as a source of energy that disrupts the sensitive energy balance in stable systems. In that way it is a younger sister of photochemistry, which does the same, but uses another type of electromagnetic energy — light — as the initiator. Radiation chemistry does not deal with radioactive elements (as radiochemistry does), except to use them as a source of radiation, always physically separated from the irradiated system.

<u>Origins and development</u>

The origins of radiation chemistry, as a new science, can be traced back to the end of the 19th century, just a few years after the discovery of X-rays. It is not surprising that chemical effects of X-rays were observed so soon, if one recalls that even soft X-rays (low energy, 100-200 kiloelectron-volts) are made of quanta having the capacity to excite or ionize atoms and molecules and break tens of thousands of molecular bonds. Early attempts to understand the nature of chemical reactions induced by radiation were hindered by the lack of sufficiently strong radiation sources to produce changes that could be measured by what, in retrospect, were very rough and insensitive analytical techniques.

Radiation chemistry

Radiation chemists are interested in different types of radiations: • electromagnetic (X-rays, gamma rays) • charged particles (including electrons, positrons, protons, heavy ions) • neutral particles (neutrons). Energies of these radiations are usually of the order of kiloelectron volts (KeV = 103 eV) or millions of electron volts (MeV * 106 eV). These are many orders of magnitude higher than the ionization and excitation energies of electrons in atoms and molecules {tens of eV), or bonding energies of atoms in molecules (several eV). One particle, or quantum, of electromagnetic radiation can ionize or excite a multitude of molecules — hence, the high efficiency of conversion of radiation energy into chemical effects. Most radiation chemical experiments are done using gamma rays (from the isotope cobalt-60) or high-energy electrons (from electron beam accelerators with energies of several MeV). The same types of radiations are used in industrial applications. Interacting with matter, these types of radiations dislocate only orbital electrons and produce excited atomic species, free radicals, and ions. These radiation energies do not induce radioactivity in matter exposed to them. Lately, interest has grown in the use of neutron beams, heavy ion beams, and electrons of very low energy (much less than 0.1 MeV).

Types of Radiation:

1. Alpha (α) particles
2. Beta (β) particles
3. Gamma (γ) rays
4. X-rays
5. Ultraviolet (UV) radiation
6. Cosmic rays

Radiation-Induced Chemical Changes:

1. Ionization: formation of ions and free radicals

2. Excitation: energy transfer to molecules
3. Dissociation: bond breaking
4. Recombination: formation of new bonds

Radiation Chemistry Applications:

1. Radiation Processing: sterilization, cross-linking, and degradation
2. Nuclear Power: radiation effects on materials and coolants
3. Space Exploration: radiation protection and effects on materials
4. Medical Applications: cancer treatment, radiation therapy
5. Environmental Remediation: radiation-induced degradation of pollutants

Key Concepts:

1. Radiation Chemical Yield (G-value)
2. Radiation Dose (Gy)
3. Linear Energy Transfer (LET)
4. Radiation Sensitivity (G(S))
5. Free Radical Chemistry

Radiation Chemistry in Various Fields:

1. Nuclear Chemistry
2. Materials Science
3. Biology and Medicine
4. Environmental Science
5. Aerospace Engineering

Notable Scientists:

1. Marie Curie

2. Ernest Rutherford
3. Niels Bohr
4. Linus Pauling
5. Alan Carrington

Radiation Chemistry Techniques:

1. Pulse Radiolysis
2. Gamma Radiolysis
3. Electron Paramagnetic Resonance (EPR)
4. Mass Spectrometry (MS)
5. Chromatography

SPECTRUM

A band of rays of different wavelengths is known as spectrum.

Light from sun and electric bulb consists of radiations of different wavelengths.

When this light is passed through a prism, it is separated into a band of different colours.

It is because, light of longer wavelengths bend to a smaller degree while light of shorter wavelengths bend to a greater degree and thus different colours are obtained as shown in the figure.

This spectrum of white light is visible to the naked eye & in known as visible spectrum.

The rays having wavelengths below violet & above red are not visible to the naked eye. These rays form invisible spectrum.

All the radiations can be divided into various regions as shown in the fig.

Each region is characterized by the limits of wavelengths or frequencies.

TYPES OF SPECTRUM

There are two types of spectrum

1. CONTINUOUS SPECTRUM

The spectrum in which rays of different wavelengths diffuse into one another and no definite boundary can be drawn between them is called continuous spectrum.

Example

When white light is passed through prism.

It forms continuous spectrum.

2. LINE SPECTRUM (or ATOMIC SPECTRUM)

The spectrum in which dark or bright lines are separated by dark spaces is called line spectrum.

When an element or its compound is volatilized on a flame. It emits light. When this light is seen through a spectrometer, distinct lines are seen separated by dark spaces.

e.g. Line spectrum of Na consists of two yellow lines separated by a definite distance.

Similarly the spectrum of hydrogen consists of a number of lines of different colours separated by definite distances.

Atomic Spectrum can be of two types.

i. Atomic absorption spectrum

ii. Atomic emission spectrum

i. Atomic Absorption Spectrum.

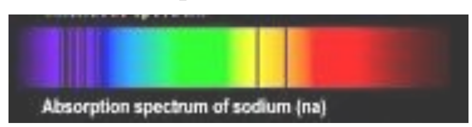
In this, dark lines are separated by bright bands.

Example.

When white light is passed through a sample of a substance it may absorb

particular radiations. This light on passing through prism will form a spectrum in which dark lines will be present in place of absorbed radiations.

It has been shown in the fig.



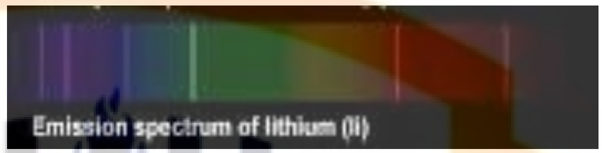
ii. Atomic Emission Spectrum.

In this, bright lines are separated by dark bands.

Example

When a substance is heated or subjected to electric discharge.

It emits radiations of definite wavelengths. These radiations will produce bright lines on a dark screen as shown in the fig.



Note:

A substance absorbs and emits radiations of same wavelengths.

Thus absorption & emission spectrum of a substance are identical in a sense that absorption lines & emission lines appears at the same place.

X-RAYS AND ATOMIC NUMBER

When fast moving electrons strike a heavy metal anode surface in a discharge tube, some highly energetic rays are produced. These rays are called X-rays.

These rays are produced when an electron suddenly stops on collision and energy is thus released in the form of X-rays. These were discovered by Roentgen.

X-rays are emitted in all directions. The wavelength of X-rays depends upon the nature of Target material. Every metal has its own characteristic X-rays.

In 1913 – 1914, an English Scientist, Mosely studied the X-rays emitted by various metals. He used 38 different elements from Al to Gold and convered a wavelength of 0.04 – 8 A0. He obtained many useful results.

Results Of Mosely's Research

1. The emitted rays are classified into two groups

a) One with shorter wavelengths are called K – series, and

b) One with longer wavelengths are called L – series.

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2. Wavelength of emitted X-rays decreases with increase in atomic number of target material.

3. The relationship between frequency (v) of a particular line in X-rays and atomic number of the element is given by

or

Where a and b are constants for a particular metal.

This linear eq is called Mosley's Law.

It states

" The frequency of X-rays emitted by an element is directly proportional to the square of atomic number of the element."

These researches show that physical and chemical properties of an element depends upon atomic number and not on atomic mass. Hence Modern periodic table is based upon atomic number.

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